Exelon Generation.

SVP-17-052

July 13, 2017 10 CFR 50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Quad Cities Nuclear Power Station, Unit 2
Renewed Facility Operating License No. DPR-30
NRC Docket No. 50-265

Subject: Licensee Event Report 265/2017-001-00, "High Pressure Coolant Injection Minimum Flow Valve Failed to Open"

Enclosed is Licensee Event Report (LER) 265/2017-001-00, "High Pressure Coolant Injection Minimum Flow Valve Failed to Open," for Quad Cities Nuclear Power Station, Unit 2.

This report is submitted in accordance with 10 CFR 50.73(a)(2)(v)(D) which requires the reporting of any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Mr. Wally J. Beck at (309) 227-2800.

Respectfully,

Kenneth S. Ohr
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

NRC FORM 366 (06-2016)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 10/31/2018

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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4. TITLE

High Pressure Coolant Injection Minimum Flow Valve Failed to Open

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
05	15	2017	2017	- 001	- 00	07	14	2017	N/A	N/A
									FACILITY NAME N/A	DOCKET NUMBER N/A

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
100	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Mark Humphrey – Regulatory Assurance	(309) 227-2810

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	BJ	FIS	B080	Y					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO		N/A	N/A	N/A

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On May 15, 2017, at 19:18 hours, station Operations personnel were performing a High Pressure Coolant Injection (HPCI) Pump Operability Test which ensures the HPCI Minimum Flow Valve opens as pump flow decreases. When the HPCI Turbine was tripped, the Minimum Flow Valve did not open as expected when system flow was reduced to the low flow setpoint. Operators took steps to open the valve manually, but upon release of the control switch, the valve returned to the closed position. The valve was then left in the closed position. The HPCI system was declared inoperable and Technical Specification 3.5.1 Condition G was entered.

The cause of the Minimum Flow Valve failing to open was attributed to the HPCI Pump Discharge Flow Indicating Switch, specifically, intermittent failure of the high side micro switch caused by residual material from the manufacturing process. The Flow Indicating Switch, which had been installed for three months, was replaced and the HPCI Pump Operability Test was successfully re-performed. The failed switch was then sent to Exelon's Power Labs for failure analysis.

The safety significance of this event was minimal. Given the impact on the HPCI system, this report is submitted for Unit 2 in accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), which requires the reporting of any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. The HPCI system is a single train system and the loss of HPCI could impact the plant's ability to mitigate the consequences of an accident.

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Quad Cities Nuclear Power Station Unit 2	05000265	2017	- 001	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

The High Pressure Coolant Injection (HPCI) [BJ] Minimum Flow Valve failed to open during HPCI Pump operability testing.

A. CONDITION PRIOR TO EVENT

Unit: 2	Event Date: May 15, 2017	Event Time: 19:18 hours
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100%

B. DESCRIPTION OF EVENT

On May 15, 2017 at 19:18 Central, station Operations personnel were performing Procedure QCOS 2300-05, "HPCI Pump Operability Test," which ensures the Motor Operated (MO) 2-2301-14, "HPCI Minimum Flow Valve," opens as pump flow decreases. During the test, the valve unexpectedly failed to open. Operators took steps to open the valve manually, but upon release of the control switch, the valve immediately returned to the closed position. Operators left the valve in the closed position. The Unit 2 HPCI system remained available but was declared inoperable.

Troubleshooting determined that the failure was due to a failure of the commercially dedicated HPCI Flow Indicating Switch (FIS) 2-2354. FIS 2-2354 had been recently replaced on February 2, 2017, under a corrective work order utilizing a commercially dedicated switch. The FIS was bench tested and post maintenance tested satisfactorily on February 2, 2017. The switch was again calibrated under a preventative maintenance work order during the next quarterly system work window on May 15, 2017, in accordance with a 92-day preventative maintenance frequency. All results were satisfactory. The switch then failed during the subsequent quarterly system run on the same day. In the follow-up failure analysis Power Labs reported that the switch had an intermittent failure of the high side micro switch due to residual material remaining from the manufacturing process.

On May 15, 2017, at 23:27 Central, ENS #52758 was made to the NRC under 10 CFR 50.72(b)(3)(v)(D), to report this event as an event or condition that could have prevented the fulfillment of a safety function.

Given the impact on the HPCI system, this report is submitted for Unit 2 in accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), which requires the reporting of any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

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C. CAUSE OF EVENT

The cause was determined to be failure of the HPCI Pump Discharge Flow Indicating Switch, specifically intermittent failure of the high side micro switch caused by residual material from the manufacturing process.

D. SAFETY ANALYSIS

System Design

According to the Quad Cities Nuclear Power Station (QCNPS), Units 1 and 2 Updated Final Safety Analysis Report (UFSAR) Section 6.3.2.3, the HPCI subsystem is designed to pump water into the reactor vessel under Loss of Coolant Accident (LOCA) conditions which do not result in rapid depressurization of the pressure vessel. The loss of coolant might be due to a loss of reactor feedwater or to a small line break which does not cause immediate depressurization of the reactor vessel. The sizing of the HPCI subsystem is based upon providing adequate core cooling during the time that the pressure in the reactor vessel decreases to a value that the Core Spray [BM] subsystem and/or the Low Pressure Coolant Injection (LPCI) [BO] subsystem become effective. The HPCI subsystem is designed to pump 5600 gallons per minute into the reactor vessel within a reactor pressure range of about 1120 pounds per square inch gage (psig) to 150 psig. Initiation of the HPCI subsystem occurs automatically on signals indicating reactor low-low water level or high drywell pressure. HPCI injection into the reactor vessel may be accomplished manually by the operator or without operator action by the HPCI automatic initiation circuitry. HPCI can also operate in a pressure control mode of consuming steam from the reactor vessel without providing full injection into the vessel (down to and including zero injection).

Safety Impact

The safety impact of this condition was minimal. Valve MO 2-2301-14 is a normally closed valve that is used to maintain minimum flow through the HPCI pump to the Suppression Pool when the injection isolation valves are shut. The bypass valve is automatically opened on low pump flow and closed on high flow whenever the steam supply valve to the HPCI turbine is open. According to UFSAR Section 6.3.2.3.3, the HPCI minimum flow system is provided for pump protection. The minimum flow valve is automatically opened on low pump flow and closed on high flow whenever the steam supply valve to the turbine is open. Even though the minimum flow valve failed to open, the HPCI System remained capable of performing its intended design/safety function and would not have hindered the system from fulfilling any required safety function or injection over the required 10 minute mission time.

According to UFSAR Table 6.2-7, MO 2-2301-14 is considered a primary containment isolation valve [ISV] with a normal position of closed. MO 2-2301-14 is the only primary containment isolation valve present in line 2-2340-4"-DX. The failure of the FIS caused MO 2-2301-14 to fail in and remain in the closed position. Since the line could effectively be isolated utilizing the primary containment isolation valve, the primary containment integrity could be assured, therefore, the primary containment system remained capable of performing its intended design/safety function.

With MO 2-2301-14 failing to open, HPCI was declared inoperable and TS 3.5.1 Condition G was entered. Required Action G.2 is to restore HPCI System to OPERABLE status, with a completion time of 14 days. The FIS was replaced, post maintenance tested and the HPCI system declared operable within the 14 day completion time. There were no other issues or problems identified with valve MO 2-2301-14 after replacement of the FIS. No other repairs were required or performed.

Since HPCI is a single train safety system, this notification is being made in accordance with 10 CFR 50.73(a)(2)(v)(D) (Event or Condition that Could Have Prevented Fulfillment of a Safety Function).

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The engineering analysis that was performed demonstrated this event did not constitute a Safety System Functional Failure (SSFF). (Reference NEI 99-02, Revision 7, Regulatory Assessment Performance Indicator Guideline, Section 2.2, “Mitigating Systems Cornerstone, Safety System Functional Failures, Clarifying Notes, Engineering Analyses.”) As such, this event will not be reported in the NRC Performance Indicator (PI) for SSFF since an engineering analysis was performed which determined that the system was capable of performing its safety function during this event.

Risk Insights

The plant Probabilistic Risk Assessment (PRA) model gives no credit for failure of the HPCI Minimum Flow Valve failure to open and does not include it in the model; hence, the failure of the valve to open did not contribute to an increase in risk.

In conclusion, due to the fact that the HPCI system was available to perform its safety function, the overall safety significance and impact on risk of this event were minimal.

E. CORRECTIVE ACTIONS

Immediate:

1. The failed FIS was replaced with a new commercially dedicated and bench tested FIS and post maintenance tested satisfactorily.

Follow-up:

1. Site Procurement will purchase the FIS from an approved Appendix B program vendor rather than purchasing a commercially dedicated part.
2. The Calibration Data Sheet for the FIS will be revised to include cycling new switches ten (10) times during bench testing.

F. PREVIOUS OCCURRENCES

The station events database, LERs, and INPO Consolidated Event System (ICES) were reviewed for similar events at QCNPS. This event was caused by an intermittent failure of the high side micro switch of the HPCI FIS which contained residual material from the manufacturing process.

No previous occurrences were identified as applicable to the circumstances of this event.

G. COMPONENT FAILURE DATA

Failed Equipment: Flow Indicating Switch
Component Manufacturer: Barton
Component Model Number: 289A
Component Part Number: N/A

This event has been reported to ICES as Report No. 415432